SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

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Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

● The team needs to embed a promotional video on the homepage.

● A registration form must securely send user data to the server.

● The navigation menu must follow semantic HTML standards.

● XML configuration files are being used but contain syntax errors.

● The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>

B. <video src="video.mp4" controls></video>

C. <movie src="video.mp4"></movie>

D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET

B. POST

C. PUT

D. DELETE

Snippet 3: Navigation Elements

A.

B. <nav>

C. <section>

D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>

B. <note><to>User</to><from>Admin</from></note>

C. <note><to>User</from></note>

D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP

B. HTTP

C. SMTP

D. SNMP

Questions:

1. Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: Option B – <video src="video.mp4" controls></video>

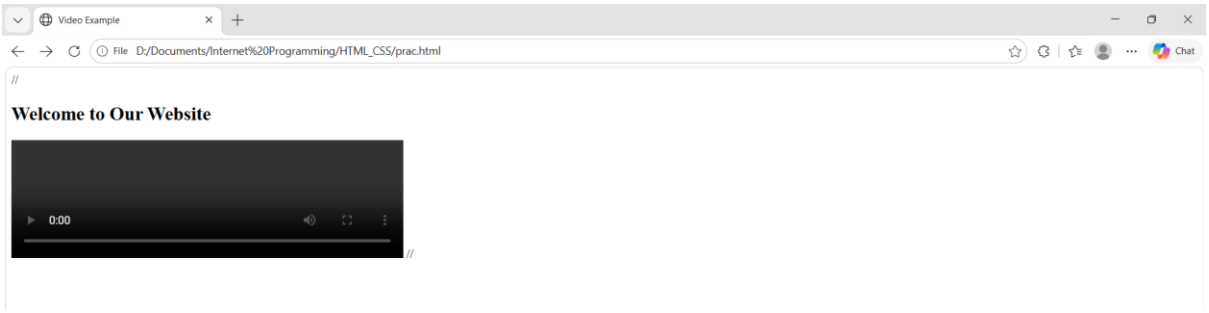
Explanation:

The HTML <video> tag is used to display video content directly on a web page without requiring additional software or external plugins.

The src attribute specifies the location of the video file, while the controls attribute provides built-in playback options such as play, pause, volume adjustment, and full screen mode.

Example:

```
<!DOCTYPE html>
<html>
<head>
   <title>Video Example</title>
</head>
<body>
   <h2>Welcome to Our Website</h2>
   <video src="video.mp4" controls width="500">
   </video>
</body>
</html>
```



Why this option is correct:

- It uses the standard HTML5 video element.
- It allows videos to be played directly in the browser.
- It provides user-friendly playback controls.
- It supports modern web development standards.

2. Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: Option B – POST

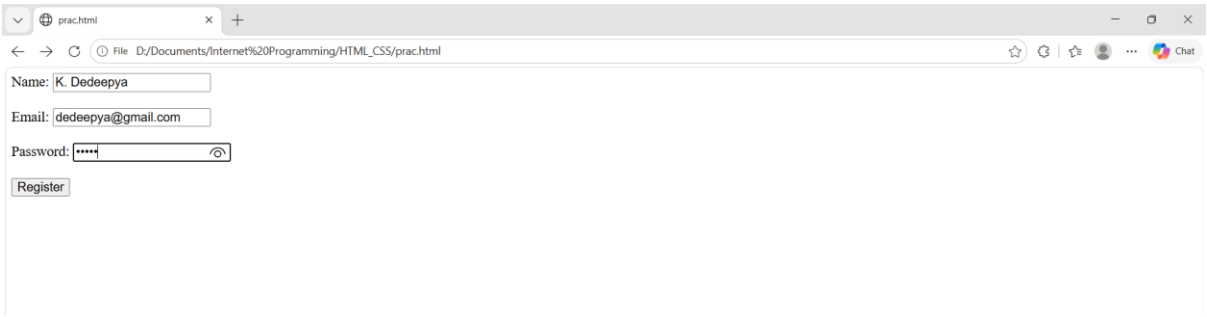
Explanation:

The POST method is commonly used to submit registration forms and transfer user information from a webpage to a web server.

Unlike the GET method, POST sends the submitted form data in the body of the HTTP request instead of displaying it in the URL. This makes it more suitable for sending information such as usernames, email addresses, and passwords.

Example:

```
<form action="register.php" method="POST">
   <label>Name:</label>
   <input type="text" name="username">
   <br><br>
   <label>Email:</label>
   <input type="email" name="email">
   <br><br>
   <label>Password:</label>
   <input type="password" name="password">
   <br><br>
   <button type="submit">Register</button>
</form>
```



Why this option is correct:

- POST transfers form information through the request body.
- It is suitable for registration and login forms.
- It can handle larger amounts of submitted data than GET.
- It does not expose submitted values directly in the URL**.**

Important security point: POST alone does not encrypt information. HTTPS must be used along with POST to protect sensitive user data during transmission.

3. Choose the proper semantic element for navigation and justify its usage.

Correct Answer: Option B – <nav>

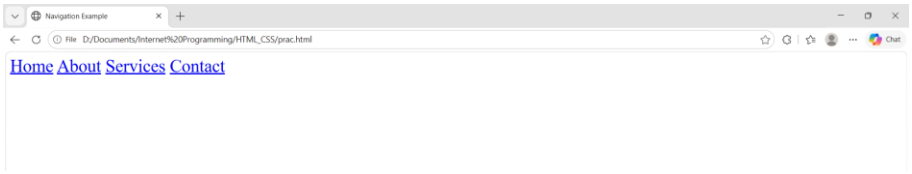
Explanation:

The HTML <nav> element is a semantic element used to define the main navigation links of a website.

It helps users move between different sections or pages, such as Home, About, Services, and Contact. It also helps browsers, search engines, and screen readers understand the purpose of the navigation section.

Example:

```
<!DOCTYPE html>
<html>
<head>
   <title>Navigation Example</title>
</head>
<body>
   <nav>
      <a href="home.html">Home</a>
      <a href="about.html">About</a>
      <a href="services.html">Services</a>
      <a href="contact.html">Contact</a>
   </nav>
</body>
</html>
```



Why this option is correct:

- It gives a clear meaning to the navigation section.
- It improves website structure and readability.
- It supports accessibility for screen reader users.
- It helps search engines understand important navigation links.

4. Identify the correct XML syntax and explain the error in other options.

Correct Answer: Option B – <note><to>User</to><from>Admin</from></note>

Explanation:

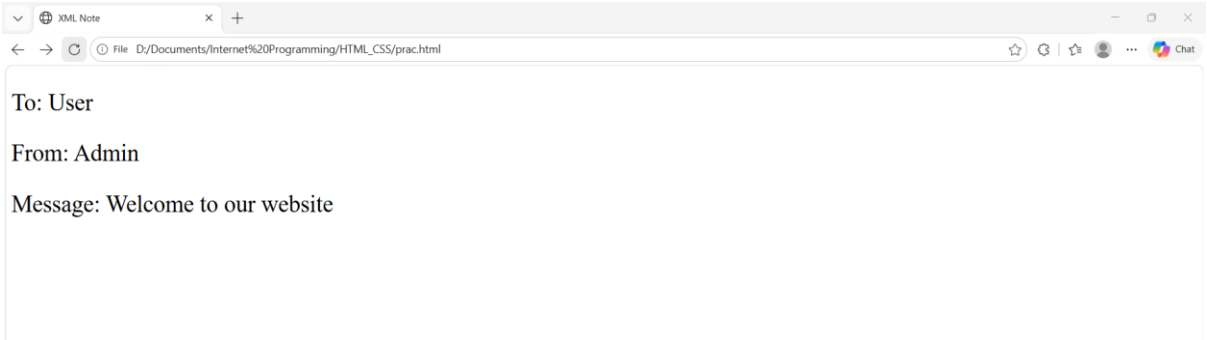
XML (Extensible Markup Language) is used to store and organize data in a structured format. XML follows strict syntax rules, where every opened element must have a corresponding closing element.

Option B is correct because all the opening and closing tags are properly written and arranged in the correct order.

Correct XML Example:

```
<?xml version="1.0"?>

<note>
   <to>User</to>
   <from>Admin</from>
   <message>Welcome to our website</message>
</note>
```



Explanation of the incorrect options:

Option	Error
A	The <note> element is not closed properly.

B	**All elements have matching opening and closing tags.**
C	**The <to> opening tag is incorrectly closed with </from>.**
D	**The final closing angle bracket > is missing.**

Why this option is correct:

- Every opening tag has a matching closing tag.
- Elements are properly nested.
- The XML document follows well-formed syntax rules.
- The structure makes information easy to read and process.

5. Choose the correct protocol for web communication and justify your choice.

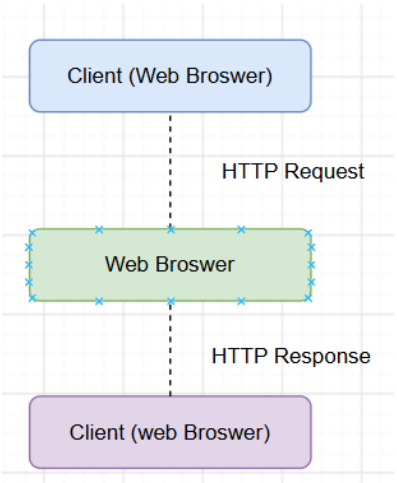
Correct Answer: Option B – HTTP (Hypertext Transfer Protocol)

Explanation:

HTTP is an application-layer protocol used for communication between web browsers and web servers. It allows browsers to request web pages, images, videos, and other resources from servers.

When a user enters a website address in a browser, HTTP enables the exchange of requests and responses between the client and server.

Example:



Why this option is correct:

- HTTP enables communication between browsers and web servers.
- It transfers HTML pages and other web resources.
- It follows a request-response communication model.
- It supports the delivery of website content over the internet.

Other options:

- **FTP:** Used for transferring files between computers.
- **SMTP:** Used for sending email messages.
- **SNMP:** Used for monitoring and managing network devices.

Security note: HTTPS is the secure version of HTTP. It uses encryption to protect communication between the browser and server.